



Figure 1: Dimensions of Web page optimisations

resolution attempts.

- Reducing the redirects can increase speed. These redirects are performed with 301 and 302 status codes.
- With respect to Web 2.0 applications, caching of AJAX (Asynchronous JavaScript And XML) requests is an important step.
- The number of DOM (Document Object Model) elements should be kept under control.

Server optimisation

- Using a Content Delivery Network (CDN) can help in optimising the Web page's performance. Geographical proximity to the user has a positive impact on the time required to fetch content.
- A cache-control header can help. If the content is static, then the expiry should be set as *Never Expire*. For dynamic content, the time up to when the component is valid should be set. This will minimise HTTP requests.
- Compressing the components is another great step in optimisation. This can be achieved with 'Gzip'. Experts estimate that the compression minimises the time required for responses by 70 per cent.
- With respect to AJAX applications, the GET method is preferable. So, along with *XMLHttpRequest*, as far as possible use the GET method.

Cookies

Cookies are one of the most used mechanisms by Web developers to store tiny pieces of information. With respect to cookies, the following factors should be considered:

- Size of the cookies should be kept minimal.
- Cookies should be set at the appropriate level in the domain hierarchy. This is done to reduce the impact on sub-domains.
- Don't forget to set a proper expiry date for the cookie.

Style sheets

Professionally designed style sheets make Web pages look elegant. The following factors must be considered in handling style sheets:

- It is better to keep the style sheets in the HEAD section of the Web pages. This is done to permit the pages to render incrementally.
- Care should be taken to use expressions in CSS.

Mathematical expressions in CSS are evaluated a lot more times than the developer might actually expect. Avoid them as far as possible.

- If you have to include multiple CSS files, merge them all into one file. This reduces the number of HTTP requests. For example, instead of the following code...

```
<link rel="stylesheet" href="1.css" media="all">
<link rel="stylesheet" href="2.css" media="all">
<link rel="stylesheet" href="3.css" media="all">
<link rel="stylesheet" href="4.css" media="all">
<link rel="stylesheet" href="5.css" media="all">
```

...use the command given below:

```
<link rel="stylesheet" href="complete.css" media="all">
```

- Opt to use `<link>` over the `@import` when using CSS in a page.

JavaScript

JavaScript has become the de-facto client-side scripting language. So the way in which JavaScript components are built does have a significant impact on the performance of Web pages.

- If possible, move the script to the bottom of the page. This cannot be done always (for example, if your page's critical contents are rendered through the *document.write()* function).
- Using external JavaScript and style sheet files will enable better caching. So, it would be better in many scenarios to put CSS and JavaScript through the external mode.
- Minifying* and *Obfuscation* are two effective mechanisms to improve the performance by tweaking the code. One survey indicates that obfuscation can achieve a 25 per cent reduction in size.
- Crowding of events needs to be avoided. Delegating events properly improves the performance of the page.
- The usage of *async* (asynchronous) must be encouraged, as shown below:

```
<script async src="example.js"></script>
```

If you don't use the *async* keyword then the page has to wait till the *example.js* is fully downloaded. The *async* keyword makes page parsing happen even before the downloading of the script is completed. Once the script is downloaded, it is activated. However, when using multiple *async*, the order of execution becomes a concern.

Optimising images

Images are an integral part of most Web pages. Hence, the way images are handled defines the performance of the application. The following factors should be considered: